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SE202L – Development Operations Lab

Lab 06 – Report **(Git Basics & API Testing)**

Lab Assessment Tasks

Task 1: Git Project Setup

```
sheikh@Ubuntu:~$ mkdir Lab#5_DevOps
sheikh@Ubuntu:~$ cd Lab#5_DevOps
sheikh@Ubuntu:~/Lab#5_DevOps$ mkdir devops_scripts
sheikh@Ubuntu:~/Lab#5_DevOps$ cd devops_scripts
sheikh@Ubuntu:~/Lab#5_DevOps/devops_scripts$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/sheikh/Lab#5_DevOps/devops_scripts/.git/
sheikh@Ubuntu:~/Lab#5_DevOps/devops_scripts$ touch README.md
sheikh@Ubuntu:~/Lab#5_DevOps/devops_scripts$ nano README.md
sheikh@Ubuntu:~/Lab#5_DevOps/devops_scripts$ cat README.md
# DevOps Scripts

This repository contains basic shell scripts created during DevOps practice.

Scripts included:
- System information script
- Backup script
- User input script
```

```
sheikh@Ubuntu:~/Lab#5_DevOps/devops_scripts$ git add README.md
sheikh@Ubuntu:~/Lab#5_DevOps/devops_scripts$ git commit -m "Add README file"
[master (root-commit) 6eb2965] Add README file
 1 file changed, 8 insertions(+)
 create mode 100644 README.md
sheikh@Ubuntu:~/Lab#5_DevOps/devops_scripts$ git touch system-info.sh
git: 'touch' is not a git command. See 'git --help'.
sheikh@Ubuntu:~/Lab#5_DevOps/devops_scripts$ touch system-info.sh
sheikh@Ubuntu:~/Lab#5_DevOps/devops_scripts$ nano system-info.sh
sheikh@Ubuntu:~/Lab#5_DevOps/devops_scripts$ git add system-info.sh
sheikh@Ubuntu:~/Lab#5_DevOps/devops_scripts$ git commit -m "Add system info script"
[master c07ac4d] Add system info script
 1 file changed, 6 insertions(+)
 create mode 100644 system-info.sh
```

```
sheikh@Ubuntu:~/Lab#5_DevOps/devops_scripts$ touch backup.sh
sheikh@Ubuntu:~/Lab#5_DevOps/devops_scripts$ nano backup.sh
sheikh@Ubuntu:~/Lab#5_DevOps/devops_scripts$ git add backup.sh
sheikh@Ubuntu:~/Lab#5_DevOps/devops_scripts$ git commit -m "Add backup script"
[master 08aa9f2] Add backup script
 1 file changed, 6 insertions(+)
 create mode 100644 backup.sh
sheikh@Ubuntu:~/Lab#5_DevOps/devops_scripts$ touch user-input.sh
sheikh@Ubuntu:~/Lab#5_DevOps/devops_scripts$ nano user-input.sh
sheikh@Ubuntu:~/Lab#5_DevOps/devops_scripts$ git add user-input.sh
```

```
sheikh@Ubuntu:~/Lab#5_DevOps/devops_scripts$ git commit -m "Add user input script"
[master fdcac76] Add user input script
 1 file changed, 6 insertions(+)
 create mode 100644 user-input.sh
sheikh@Ubuntu:~/Lab#5_DevOps/devops_scripts$ touch .gitignore
sheikh@Ubuntu:~/Lab#5_DevOps/devops_scripts$ nano .gitignore
sheikh@Ubuntu:~/Lab#5_DevOps/devops_scripts$ git add .gitignore
sheikh@Ubuntu:~/Lab#5_DevOps/devops_scripts$ git commit -m "Add gitignore file"
[master c997330] Add gitignore file
 1 file changed, 3 insertions(+)
 create mode 100644 .gitignore
```

```
sheikh@Ubuntu:~/Lab#5_DevOps/devops_scripts$ git log --oneline --graph > git-history.txt
sheikh@Ubuntu:~/Lab#5_DevOps/devops_scripts$ cat git-history.txt
* c997330 Add gitignore file
* fdcac76 Add user input script
* 08aa9f2 Add backup script
* c07ac4d Add system info script
* 6eb2965 Add README file
```

Task 2: Git Workflow Documentation

1. Purpose of Git

Git is a version control system used to track changes in files and manage project versions. It helps developers collaborate and maintain code history.

2. Git Commands

- git init – Initialize a Git repository
- git add – Add files to staging area
- git commit – Save changes to repository
- git status – Show file status
- git log – Show commit history
- git log --oneline – Short commit history
- git diff – Show file changes
- git branch – List or create branches
- git checkout – Switch branches
- git push – Upload commits to remote repository

3. Three Git Areas

- Working Directory – Where files are edited
- Staging Area – Files prepared for commit using `git add`
- Repository – Final saved versions using `git commit`

4. Good Commit Messages

- Add README file
- Fix login bug
- Update backup script
- Add user input script

5. When to Use .gitignore

`.gitignore` is used to ignore files that should not be tracked by Git such as:

API TESTING USING CURL:

```
C:\Users\sheik>curl -X POST http://127.0.0.1:5000/register ^
More? -H "Content-Type: application/json" ^
More? -d "{\"full_name\":\"Sheikh Nabeel\", \"username\":\"nabeel8090\", \"password\":\"sheikhnabeel8090\"}"
{
  "message": "User nabeel8090 registered successfully."
}

C:\Users\sheik>curl -X POST http://127.0.0.1:5000/login ^
More? -H "Content-Type: application/json" ^
More? -d "{\"username\":\"nabeel8090\", \"password\":\"sheikhnabeel8090\"}" ^
More? -c cookies.txt
{
  "message": "User nabeel8090 logged in successfully."
}

C:\Users\sheik>curl http://127.0.0.1:5000/dashboard -b cookies.txt
{
  "full_name": "Sheikh Nabeel",
  "username": "nabeel8090"
}

C:\Users\sheik>curl http://127.0.0.1:5000/profile/nabeel8090
{
  "full_name": "Sheikh Nabeel",
  "username": "nabeel8090"
}

C:\Users\sheik>curl -X POST http://127.0.0.1:5000/logout -b cookies.txt
{
  "message": "Logged out successfully."
}
```

== THE END ==